discussion session: Apr. 8, 2024 ☆ There are some extra problems

textbook: 11.8 Power Series

"power series about a": $\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$ (constants (coefficients))

3 possibilities:

① converges only if $x = a$ → (trivial case: $\sum 0 = 0$) (see ex 2)

② converges for all x (see ex 3)

③ there is a positive real number R such that $|x-a| < R \Rightarrow$ converges, $|x-a| > R \Rightarrow$ diverges

↳ R = radius of convergence (use ratio test or root test to get R)

Check endpoints $a-R$ and $a+R$ to get interval of convergence.

$\boxed{\text{ex 1}}$ $\sum_{n=1}^{\infty} \frac{(3x)^n}{(n+1)^{3/2}}$

$a_n = \frac{(3x)^n}{(n+1)^{3/2}}$

$a_{n+1} = \frac{(3x)^{n+1}}{(n+2)^{3/2}} = \frac{(3x)^n(3x)}{(n+2)^{3/2}}$

ratio test

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(3x)^n(3x)}{(n+2)^{3/2}}\cdot\frac{(n+1)^{3/2}}{(3x)^n}\right| = \lim_{n\to\infty}\left|\frac{3x\,(n+1)^{3/2}}{(n+2)^{3/2}}\right|$$

$$= \lim_{n\to\infty}|3x|\left|\frac{n+1}{n+2}\right|^{3/2} = |3x|\lim_{n\to\infty}\left(\frac{n+1}{n+2}\right)^{3/2}$$

↖ if n large, $\approx \frac{n}{n} = 1$

$$= |3x| = 3|x| < 1 \Rightarrow |x| < \frac{1}{3}$$

$R = \frac{1}{3}$

$R = \frac{1}{3}$. $|x| < \frac{1}{3} \Rightarrow -\frac{1}{3} < x < \frac{1}{3}$

Check endpoints:

$x = -\frac{1}{3}$: $\sum_{n=1}^{\infty} \frac{(3\cdot(-\frac{1}{3}))^n}{(n+1)^{3/2}} = \sum_{n=1}^{\infty} \frac{(-1)^n}{(n+1)^{3/2}}$

AST: $b_n = \frac{1}{(n+1)^{3/2}}$ ① is positive for all n ② decreasing

③ $\lim_{n\to\infty} b_n = 0$ so converges for $x = -\frac{1}{3}$

$x = \frac{1}{3}$: $\sum_{n=1}^{\infty} \frac{(3\cdot\frac{1}{3})^n}{(n+1)^{3/2}} = \sum_{n=1}^{\infty} \frac{1}{(n+1)^{3/2}}$ compare to $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ ← convergent p-series ($p = \frac{3}{2} > 1$)

$\frac{1}{(n+1)^{3/2}} < \frac{1}{n^{3/2}} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{(n+1)^{3/2}}$ converges by Comparison Test.

So, interval of convergence is $\left[-\frac{1}{3}, \frac{1}{3}\right]$

$\boxed{\text{ex 2}}$ $\sum_{n=1}^{\infty} \frac{(x+5)^n (n+5)!}{n^5}$

$a_{n+1} = \frac{(x+5)^{n+1}(n+6)!}{(n+1)^5} = \frac{(x+5)^n(x+5)(n+6)(n+5)!}{(n+1)^5}$

ratio test

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(x+5)^n(x+5)(n+6)(n+5)!}{(n+1)^5}\cdot\frac{n^5}{(x+5)^n(n+5)!}\right| = \lim_{n\to\infty}\left|\frac{(x+5)(n+6)n^5}{(n+1)^5}\right|$$

$$= \lim_{n\to\infty}|x+5|\left|\frac{(n+6)n^5}{(n+1)^5}\right| = |x+5|\lim_{n\to\infty}\frac{(n+6)n^5}{(n+1)^5} = \infty$$ diverges so $R = 0$

↖ If n large, $\approx \frac{n^6}{n^5} = n$. As $n\to\infty$, $\lim n = \infty$

and $I = \{-5\}$.

(just the point that makes $|x+5| = 0$)

$\boxed{\text{ex 3}}$ $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{2\cdot 5\cdot 8\cdots(3n-1)}$

$$a_{n+1} = \frac{x^{n+1}}{2\cdot 5\cdot 8\cdots(3n-1)(3n+2)} = \frac{x^n\, x}{2\cdot 5\cdot 8\cdots(3n-1)(3n+2)}$$

ratio test

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{x^n x}{2\cdot 5\cdot 8\cdots(3n-1)(3n+2)}\cdot\frac{2\cdot 5\cdot 8\cdots(3n-1)}{x^n}\right| = \lim_{n\to\infty}\left|\frac{x}{3n+2}\right| = 0 \Rightarrow R=\infty$$

$I = (-\infty, \infty)$

$\boxed{\text{ex 4}}$ $\displaystyle\sum_{n=1}^{\infty} 2^n n^2 x^n$

$$a_n = 2^n n^2 x^n = (2x)^n n^2$$

root test:

$$\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\sqrt[n]{|(2x)^n n^2|} = \lim_{n\to\infty}\sqrt[n]{|2x|^n n^2} = \lim_{n\to\infty}\sqrt[n]{|2x|^n}\,\sqrt[n]{n^2}$$

$$= \lim_{n\to\infty}|2x|\sqrt[n]{n^2} = |2x|\lim_{n\to\infty} n^{\frac{2}{n}} = |2x| = 2|x| < 1 \Rightarrow |x| < \frac{1}{2}$$

$R = \frac{1}{2}$

$n\to\infty$, $\frac{2}{n}\to 0$. To show: $\ln(n^{2/n})\to 0$.
Then, $e^{\ln(n^{2/n})} = n^{2/n} \to e^0 = 1$.
$\lim_{n\to\infty}(n^{2/n}) = \lim_{n\to\infty}\left(\frac{2}{n}\ln n\right) = \lim_{n\to\infty}\left(\frac{2\cdot 1/n}{1}\right) = 0$ (L'H)

Check endpoints:

$x = -\frac{1}{2}$: $\displaystyle\sum_{n=1}^{\infty}\left(2\left(-\tfrac{1}{2}\right)\right)^n n^2 = \sum_{n=1}^{\infty}(-1)^n n^2$

$\lim_{n\to\infty}(-1)^n n^2 = \infty$ diverges $\Rightarrow \sum_{n=1}^{\infty}(-1)^n n^2$ diverges by Test for Divergence

$x = \frac{1}{2}$: $\displaystyle\sum_{n=1}^{\infty}\left(2\cdot\tfrac{1}{2}\right)^n n^2 = \sum_{n=1}^{\infty} n^2$ also diverges by Test for Divergence (since $\lim_{n\to\infty} n^2 = \infty$)

So, interval of convergence is $\boxed{\left(-\frac{1}{2}, \frac{1}{2}\right)}$.

$\boxed{\text{ex 5}}$ $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n(x-3)^n}{\sqrt[3]{n}}$

root test:

$$\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\sqrt[n]{\left|\frac{(x-3)^n}{n^{1/3}}\right|} = \lim_{n\to\infty}\sqrt[n]{\frac{|x-3|^n}{n^{1/3}}} = \lim_{n\to\infty}|x-3|\frac{1}{\sqrt[n]{n^{1/3}}} = |x-3|\lim_{n\to\infty}\frac{1}{(n^{1/3})^{1/n}}$$

$= \lim_{n\to\infty} n^{-\frac{1}{3n}}$
$n\to\infty$, $-\frac{1}{3n}\to 0$.
see ex. 4.

$$= |x-3| < 1$$

$\Rightarrow R = 1$

$$-1 \overset{?}{<} x-3 \overset{?}{<} 1$$
$$\Rightarrow 2 \overset{?}{<} x \overset{?}{<} 4$$

Check endpoints:

$x = 2$: $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n(-1)^n}{\sqrt[3]{n}} = \sum_{n=1}^{\infty}\frac{1}{\sqrt[3]{n}} = \sum_{n=1}^{\infty}\frac{1}{n^{1/3}}$ ← divergent p-series ($p = \frac{1}{3} \le 1$)

$x = 4$: $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n(1)^n}{\sqrt[3]{n}} = \sum_{n=1}^{\infty}\frac{(-1)^n}{\sqrt[3]{n}}$ AST: ① $b_n = \frac{1}{\sqrt[3]{n}}$ is positive, ② decreasing, ③ $\lim_{n\to\infty} b_n = 0$ so the series converges

So, interval of convergence is $(2, 4]$

ex 6 $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2^n(n+1)}(x-6)^n$

$$a_n = \frac{n}{n+1}\left(\frac{x-6}{2}\right)^n$$

<u>root test</u>

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\frac{n}{n+1}\left(\frac{x-6}{2}\right)^n\right|} = \lim_{n\to\infty} \sqrt[n]{\left(\frac{n}{n+1}\right)} \sqrt[n]{\left|\frac{x-6}{2}\right|^n} = \frac{|x-6|}{2} \lim_{n\to\infty} \sqrt[n]{\frac{n}{n+1}}$$

$$= \frac{|x-6|}{2} \lim_{n\to\infty} \sqrt[n]{\frac{1}{1+\frac{1}{n}}} = \frac{|x-6|}{2} < 1$$

$$\Rightarrow |x-6| < 2$$

$$\Rightarrow R = 2$$

check endpoints: $-2 \overset{?}{<} x-6 \overset{?}{<} 2$

$4 \overset{?}{<} x \overset{?}{<} 8$

$x=4$: $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2^n(n+1)}(-2)^n = \sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$ (note $b_{n+1} \geq b_n$, so can't use AST)

Test for Divergence: $\displaystyle\lim_{n\to\infty} \frac{(-1)^n n}{n+1}$ does not converge. So, the series is divergent.

$x=8$: $\displaystyle\sum_{n=1}^{\infty} \frac{n(2)^n}{2^n(n+1)} = \sum_{n=1}^{\infty} \frac{n}{n+1}$

Test for Divergence: $\displaystyle\lim_{n\to\infty} \frac{n}{n+1} = \lim_{n\to\infty} \frac{1}{1+\frac{1}{n}} = 1 \neq 0$. So, the series is divergent.

So, interval of convergence is $(4, 8)$.